GE Medical Systems

Technical Publications

Direction 2135785
Revision 1

Signa® 1.0T Quadrature CTL Phased Array Coil

Operating Documentation

Language Policy For Service Documentation (Dir. 2128126)

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ZU SORGEN.

- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH NICHT ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

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- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERA DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECANICAS O DE OTRA NATURALEZA.

Language Policy For Service Documentation (Dir. 2128126) - cont'd

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONIVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A‘ CHOQUES ELÉTRICOS, MECANICOS OU OUTROS.

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- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL’APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL’ADDETTO ALLA MANUTENZIONE, ALL’UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

Language Policy For Service Documentation (Dir. 2128126) - cont'd

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Table of Contents

Table of Contents (cont'd)

Press Here to Return

Section 1 - Introduction

1-1 How the Quadrature CTL Phased Array Coil Operates

The Signa Quad CTL Phased Array Coil is designed for cervical, thoracic, and lumbar spine imaging on Signa 1.0T systems with Phased Array capability. It is a receive-only Phased Array Coil containing seven coil elements and using two, three, or four simultaneous receiver channels. The Quad Phased Array Coil uses seven signal inputs, driving seven Phased Array Input Ports. See Illustration 1-1. For installation, see Section 2-3, *Installation of Multi-Coil Preamp Assemblies for Channels 1 and 8*.

T/L Section: The T/L portion of the coil consists of four quad planar coil elements, each similar to the Quad T/L Spine Coil with Positioner, but smaller in the S-I direction. The two quad mode outputs of each coil element are combined using a +45 and -45 degree phase shifter; the four combined signals drive four separate receiver channels; four inputs. Isolation of sources for the two mode outputs from each coil element is accomplished as in a conventional quad planar coil. The four quad coil elements are isolated by positioning the overlap of the adjacent coils to cancel the coupling. With the multiple element nature of the coil elements, it is not possible to employ preamplifier decoupling utilized with other Phased Array coils on the Signa Quad CTL Phased Array Coil; the +45 and -45 degree phase shifters allow partial preamplifier decoupling.

1-1 How the Quadrature CTL Phased Array Coil Operates (cont'd)

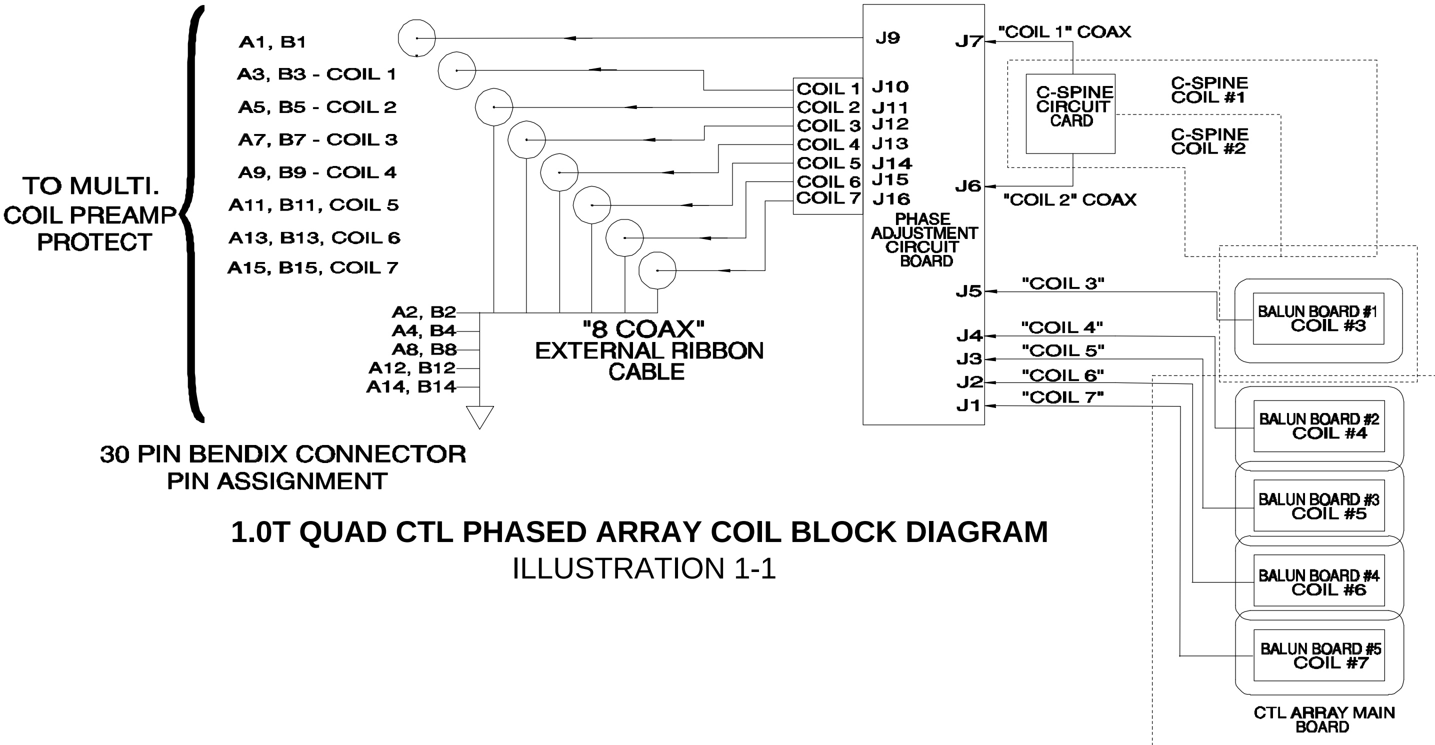
Each T/L Spine coil element consists of a single loop vertical field coil, and a two loop horizontal field coil. To resonate each coil, distributed capacitors about the loops are used. At the signal output port of each mode of the coil element, a phase shifter, one +45 degrees to a 100 ohm, 90 degree lattice balun and the other -45 degrees to a 100 ohm, 90 degree lattice balun, is used to combine the coils. The outputs are combined.

C-Spine Section: The C-Spine portion of the coil consists of three coil elements; two single loop conformal coil elements in the most superior region, one conformal quadrature element inferior to the two loops. The quad elements are schematically similar to the planar coil elements of the T/L portion of the coil. The two superior loops are similar to the Quad C-Spine Coil, but smaller in the S-I direction. The outputs of the two coil elements drive two separate input ports, to a total of two upper cervical spine inputs. The third element, inferior to the above loops, cover the lower cervical spine region, and drive a separate port. There are seven inputs; up to four coils can be selected at one time. Isolation of signal sources for the two superior coil elements is accomplished like a linear array coil. With the single-element nature of the two coil elements, the preamplifier decoupling used with other Phased Array coils can be used. Isolation for the inferior coils is the same as the T/L section.

1-1 How the Quadrature CTL Phased Array Coil Operates (cont'd)

Transmit Decoupling: As these coils are receive-only coils, decoupling from the transmit RF field is needed. This is accomplished by a passive decoupling circuit on each loop of each coil element. The decoupling network consists of a pair of high speed switching diodes in a crossed configuration in series with an inductor; these components are shunted across a series distributed capacitor in each loop. The reactance of the inductor is chosen to parallel resonate the capacitor at the operating frequency of the coil. This places a high impedance in series with each loop at the capacitor location during transmit RF pulses, thus decoupling the coil from the transmit RF field. PIN diodes are used at the feedpoints for a similar function; these diodes are biased by the system T/R Driver via the coil output cable. The external cable includes eight mating SMA plugs to connect to the phase compensation board; the cable is attached by two screws to the coil top housing. A 30 pin Bendix connector is used to mate with the system Phased Array Coil Port. The Signa 1.0T Quad CTL Phased Array Coil drives seven coil inputs, Ports 2, 3, 4, 5, 6, 7, and 8; these correspond to receiver channels 1, 2, 3, 0, 1, 2, and 3 respectively.

1-1 How the Quadrature CTL Phased Array Coil Operates (cont'd)



1.0T QUAD CTL PHASED ARRAY COIL BLOCK DIAGRAM
ILLUSTRATION 1-1

1-2 Compatibility

The 1.0T Quadrature CTL Phased Array Coil is compatible with the following hardware configurations:

- Signa® Advantage™ 1.0T System with Phased Array upgrade
- Signa® Horizon™ 1.0T System with Phased Array upgrade

1-3 Related Documents

- Direction 15400, Signa Advantage 1.5T, 1.0T, & 0.5T System
- Direction 2124201-3, MR release 5.x Signa Service Methods
- Direction 2187583-3, MR release 5.x and 8.x Signa Service Methods

1-4 Organization of this Document

This manual is divided into the following sections:

Section 1, *Introduction*, describes how the 1.0T Quad CTL Phased Array Coil operates.
Section 2, *Setup and Calibration*, describes installation procedures.
Section 3, *Functional Checks*, describes the normal power-up sequence.
Section 4, *Replacement/Maintenance*, describes field maintenance procedures.
Section 5, *Renewal Parts*, lists field replaceable parts.

1-5 Environmental Requirements

Operate and store the 1.0T Quadrature CTL Phased Array Coil in the Scanner Room.

Dimensions: 126 cm x 42.7 cm x 17.5 cm (49.6" x 16.8" x 6.9")

Section 2 - Setup and Calibration

2-1 Checking the Shipping List

Table 2-1 lists the M1887CT Signa 1.0T Quadrature CTL Phased Array Coil parts. Check that all parts have been shipped.

TABLE 2-1
1.0T QUADRATURE CTL PHASED ARRAY COIL SHIPPING LIST

QTY.	ITEM	PART NUMBER
1	1.0T Quadrature CTL Phased Array Coil	2151178-4
6	Lid Upgrade for Preamps	2148283
2	1.0T Multicoil Preamp Asm.	46-328040G6
1	Phantom Holder	2100937-16
6	100 mm Phantoms	46-317586G1
1	Operator Manual	2131388
1	Service Manual	2135785

2-2 Installing the 1.0T Quadrature CTL Phased Array Coil

At the Console, install a new softkey. This softkey will be used by the operator to select CTLTOP, CTLBOTTOM, CTLMIDDLE, CS12, CS123, TS45, LS56, and LS67.

Refer to the Signa & Signa Advantage System Manuals for information on installing soft keys (use Coil/Phased Array default values in Table 2-2 and Table 2-3).

TABLE 2-2
COIL VALUES

SYSTEM	COIL NAME	COIL TYPE	EXTREMITY COIL	CABLE LOSS	COIL LOSS	RECON SCALE FACTOR (RSF)	XMIT COIL	MULTI-COIL
1.0T	CTLTOP thru LS67	SURF.	NO	1.24	0.103 + 0.549 #	Coil RSF x 0.75	QUAD	YES

(+)= 55 cm Body Coil
(#)= 60 cm Body Coil

2-2 Installing the 1.0T Quadrature CTL Phased Array Coil (cont'd)

TABLE 2-3
PHASED ARRAY VALUES

MULTICOIL NAME	NUMBER OF RECEIVERS	START RECEIVER	STOP RECEIVER	PORT ENABLE MASK	ERROR ENABLE MASK
CTLTOP	4	0	3	7	7
CTLBOT	4	0	3	12	12
CTLMID	4	0	3	14	14
CS 12	2	1	2	3	3
CS 123 **	4	0	3	3	3
TS 45	2	0	1	4	4
LS 56	2	1	2	12	12
LS 67	2	2	3	8	8

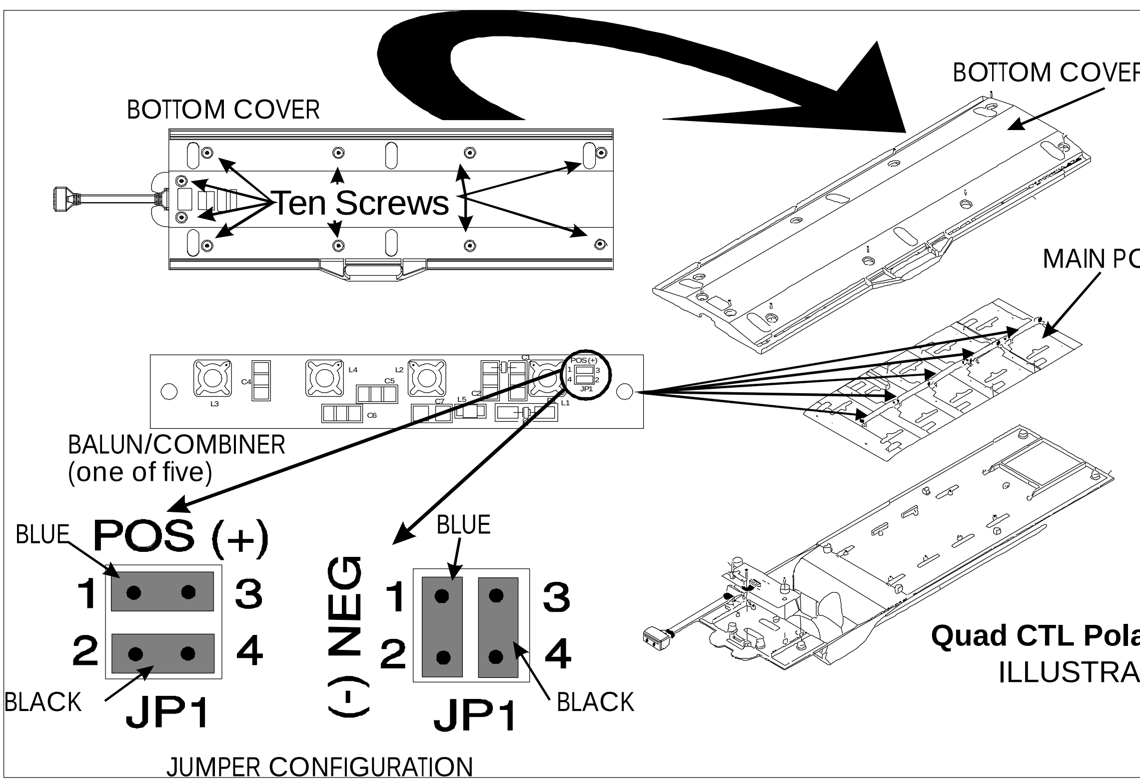
****** It is not possible to configure Signa for a three coil acquisition directly. The CS 123 is a four coil acquisition. There is no coil on Port 1; effectively, three coils are used.

2-2 Installing the 1.0T Quadrature CTL Phased Array Coil (cont'd)

Prior to use, the Quadrature polarity of the coil must be set to match the polarity of the host system. The Quad CTL Phased Array coil is shipped configured for POSITIVE or NORMAL polarity. To determine the polarity of the host system, examine the Quadrature Head Coil Quick Disconnect Box. If it is marked NORMAL, the system polarity is NORMAL or POSITIVE. If the Box is marked REVERSE, the system polarity is REVERSE or NEGATIVE. System polarity can be also verified by checking the value of CXFULL in the Configuration File. If CXFULL has a minus sign, the system polarity is REVERSE or NEGATIVE. No polarity sign indicates NORMAL or POSITIVE.

To change the polarity of the Quad CTL PA Coil, Refer to Section 4-1. Remove the bottom cover. Locate the five sets of Polarity Jumpers on the four T/L sections and the lower C-Spine section of the coil. See Illustration 2-1. For POSITIVE polarity, set the jumpers parallel to the housing-one jumper color visible. For NEGATIVE polarity, set the jumpers perpendicular to the coil housing-both jumper colors visible. All five jumper pairs are to be in the same position. Align top and bottom covers and install the ten screws.

2-2 Installing the 1.0T Quadrature CTL Phased Array Coil (cont'd)



Quad CTL Polarity Settings
ILLUSTRATION 2-1

2-3 Installation of Multi-coil Preamp Assemblies for Channel 1 & 8

1. Remove Carriage Cover Access Panel from rear of Carriage Cover.

2. Install Channel 1 upper preamplifier assembly, per Illustration 2-2.

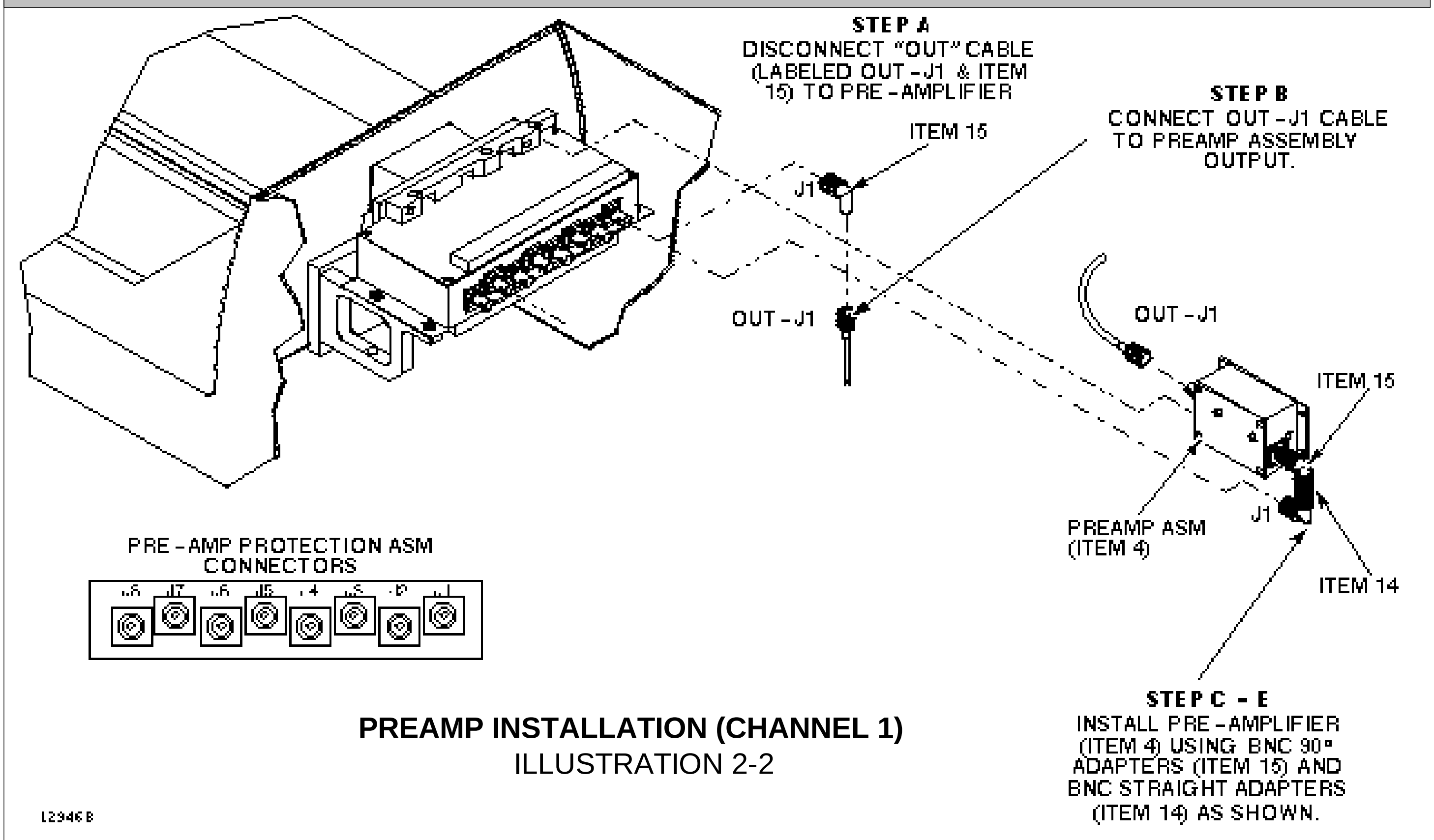
a. Disconnect output cable J1 and BNC right angle adapter (item 15) from channel 1 OUTPUT of preamp Protection Assembly. Discard BNC right angle adapter.

b. Connect output cable J1 to new preamplifier OUTPUT.

c. Insert new Preamplifier assembly into applicable notch on Preamp Protection locating block.

d. Assemble two BNC right angle connectors and straight adaptors to new Preamplifier INPUT.

e. Connect right angle adapter to J1 on Preamp Protection assembly from Preamplifier INPUT.



PREAMP INSTALLATION (CHANNEL 1)
ILLUSTRATION 2-2

2-3 Installation of Multi-coil Preamp Assemblies for Channel 1 & 8 (cont'd)

3. Install Channel 8 lower preamplifier assembly, see Illustration 2-3.

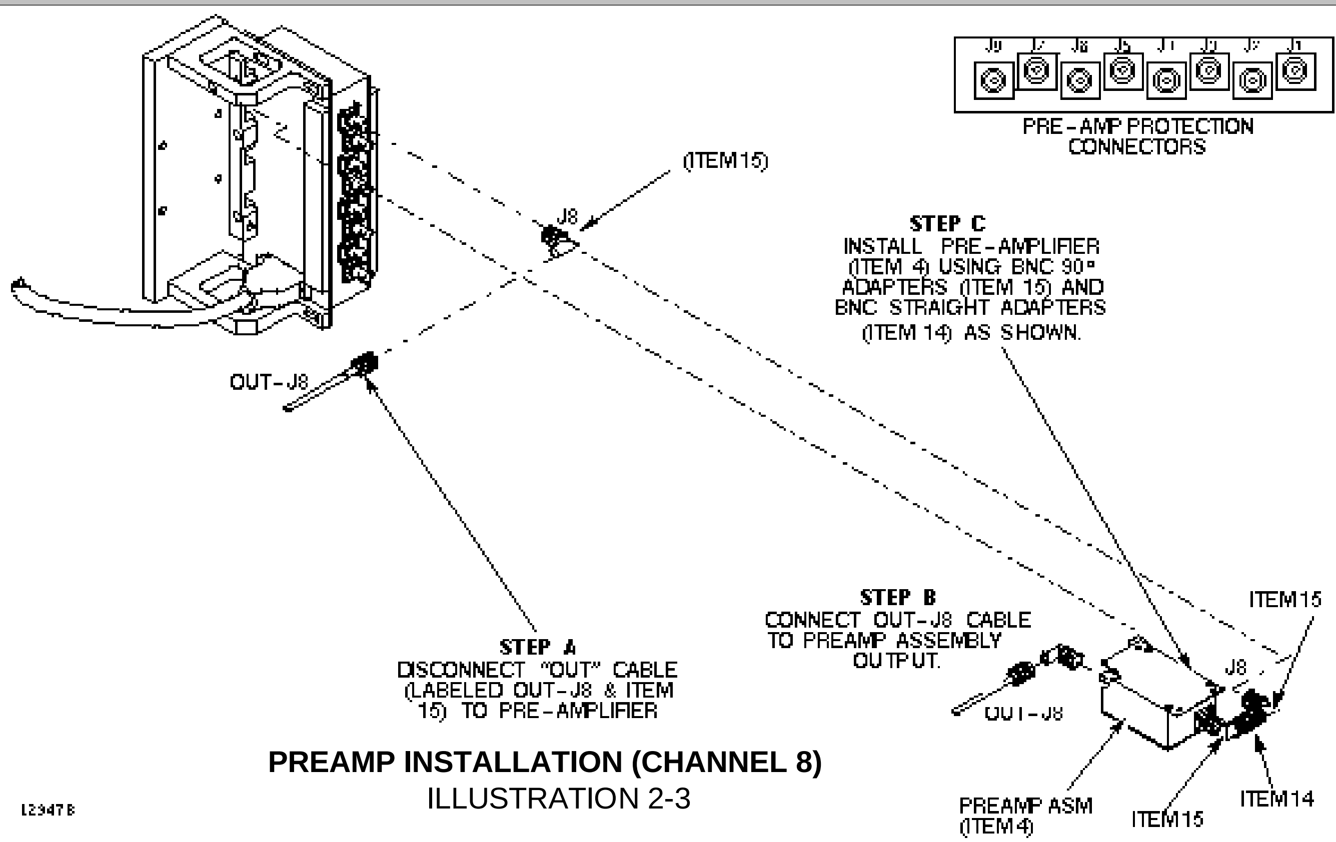
a. Disconnect output cable J8 and BNC right angle adapter (item 15) from channel 8 OUTPUT of Preamp Protection Assembly. Connect output cable J8 and BNC right angle adapter to new preamplifier OUTPUT.

b. Insert new Preamplifier assembly into applicable notch on Preamp Protection locating block.

c. Assemble two BNC right angle connectors and straight adaptors to new Preamplifier INPUT. Connect right angle adapter to J8 on Preamp Protection assembly from Preamplifier INPUT.

4. Add Foam to lid of each of the six existing Preamps, per document 2148283APR in Kit 2148283, included with the coil.

NOTE: The two preamps in Table 2-1 have been upgraded.



PREAMP INSTALLATION (CHANNEL 8)
ILLUSTRATION 2-3

2-4 Functional Checks

1. Perform a Body Coil Scan Performance Verification. Refer to Section 3-1, Signa Advantage 1.0T Scanner Verification.

2. Perform a Quad CTL Phased Array Coil Performance Verification. Refer to Section 3-2, Quadrature CTL Phased Array Coil Imaging Performance Verification.

2-5 Periodic Quality Assurance Check

On a periodic basis, such as during planned maintenance, perform the quality assurance checks outlined below to ensure that the coil is operating properly:

1. Check the external cable for cracks or breaks once a week. Refer to Section 3-4, Checking the Protectin PIN Diodes and Output cable with DMM.

2. Perform a coil SNR verification. Refer to Section 3-2, Quad CTL Phased Array Coil Imaging Performance Verification.

3. Record the date and value calculated in Section 3-3, SNR Image Analysis in column 2 or 3 under "SNR Data QA Check" of the Data Table as is instructed.

4. As is instructed in the Data Table, divide the SNR value obtained in the periodic QA check by the original SNR value and record in column 4 of the Data Table.

5. If this ratio is not greater than 85%, then there may be a problem in the Multicoil system. Contact your local GE Service Representative.

Section 3 - Functional Checks

3-1 Signa Advantage 1.0T Scanner Verification

Note

An alternate proprietary procedure is available for GE use and to customers with a valid Advanced Service Package Limited License. Refer to "TLT PROCEDURE" located on *CD-ROM Direction 15520, navigate to System: Troubleshooting*.

Phantom Required

• 6-100 mm Phantoms, 46-317586G1

3-1 Signa Advantage 1.0T Scanner Verification (cont'd)

Setup Procedure

Caution

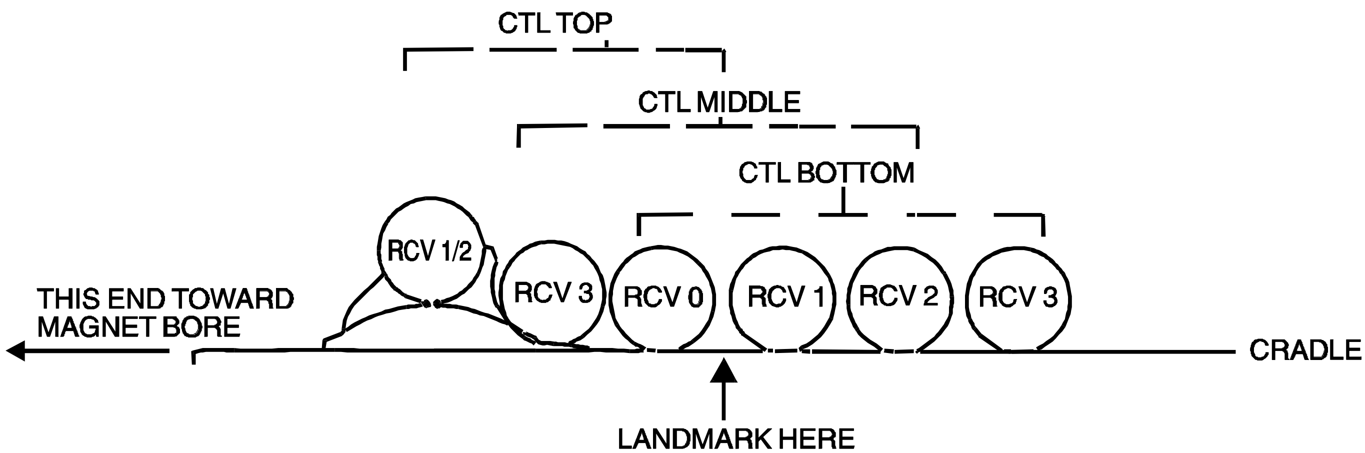
The Quad Head Coil must be completely removed from the cradle before performing any body scans. Failure to do this may result in damage to the Head Coil T/R Network.

1. Remove Quad Head Coil (if present) from cradle.

2. Select **[New Series]** to allow a new Landmark to be set.

3. Position the CTL Phantom/Positioner with phantoms in the center of the cradle. Landmark the center of the positioner with phantoms and advance to isocenter using the **[ADVANCE TO SCAN]** button. See Illustration 3-1.

3-1 Signa Advantage 1.0T Scanner Verification (cont'd)



QUAD CTL PHANTOM/POSITIONER LANDMARK SETUP
ILLUSTRATION 3-1

3-1 Signa Advantage 1.0T Scanner Verification (cont'd)

4. Setup Scan Prescription as shown in Table 3-1.

TABLE 3-1
BODY COIL SNR - SCAN PROTOCOL

Id:	**geservice**	Rep Time (TR):	**[300 msec]**
Name:	**body snr**	Auto CF:	**[Peak]**
Patient Weight:	**300**	Field of View:	**[48 cm]**
Patient Entry:	**[Head First]**	Scan Thickness:	**[3 mm]**
Patient Position:	**[Supine]**	Interscan Spacing:	**[Other] 0**
Axial/Sag. Landmark:	**[Sternal Notch]**	Start Loc (L/R): **0** End Loc (L/R): **0**	
Coil Type:	**[Body Coil]**	No. of Scan Location:	**1**
Scan Plane:	**[Sagittal]**	FOV Center (P/A):	**0** (P/A) **0**
Image Mode:	**[2D]**	Acq. Matrix (freq.):	**[256]**
Pulse Sequence:	**[Spin Echo]**	Acq. Matrix (phase)	**[256]**
Imaging Options:	**[None]**	Frequency Direction:	**[S/I]**
or enter PSD Filename		Imaging Time:	**[1 NEX 1:23]**
Number of Echoes:	**[1]**	Contrast:	**[No]**
Echo Time (TE):	**[30 msec]**	Table Delta:	**0 mm**

3-1 Signa Advantage 1.0T Scanner Verification (cont'd)

5. Select **[Auto Prescan]** to properly calibrate the RF power level for the 90 degree and 180 degree pulses.

6. Select **[Scan]**. Observe the resulting images. Ensure that there are no artifacts of any sort in the resulting image. Record the Exam number and Series number for SNR Calculations.

7. Select **[Scan]** again. This second image will be used for determination of Body Coil mode SNR.

8. Select **[Cancel]**. Refer to Section 3-3 for SNR Image Analysis.

3-2 Quadrature CTL Phased Array Coil Imaging Performance Verification

Note

An alternate proprietary procedure is available for GE use and to customers with a valid Advanced Service Package Limited License. Refer to "TLT PROCEDURE" located on *CD-ROM Direction 15520, navigate to System: Troubleshooting*.

Phantom Required

- 6-100 mm Phantoms, 46-317586G1

3-2 Quadrature CTL Phased Array Coil Imaging Performance Verification (cont'd)

Setup Procedure

1. Select **[New Series]** to allow a new Landmark to be set.

Caution

The Quad Head Coil must be completely removed from the cradle before performing any body scans. Failure to do this may result in damage to the Head Coil T/R Network.

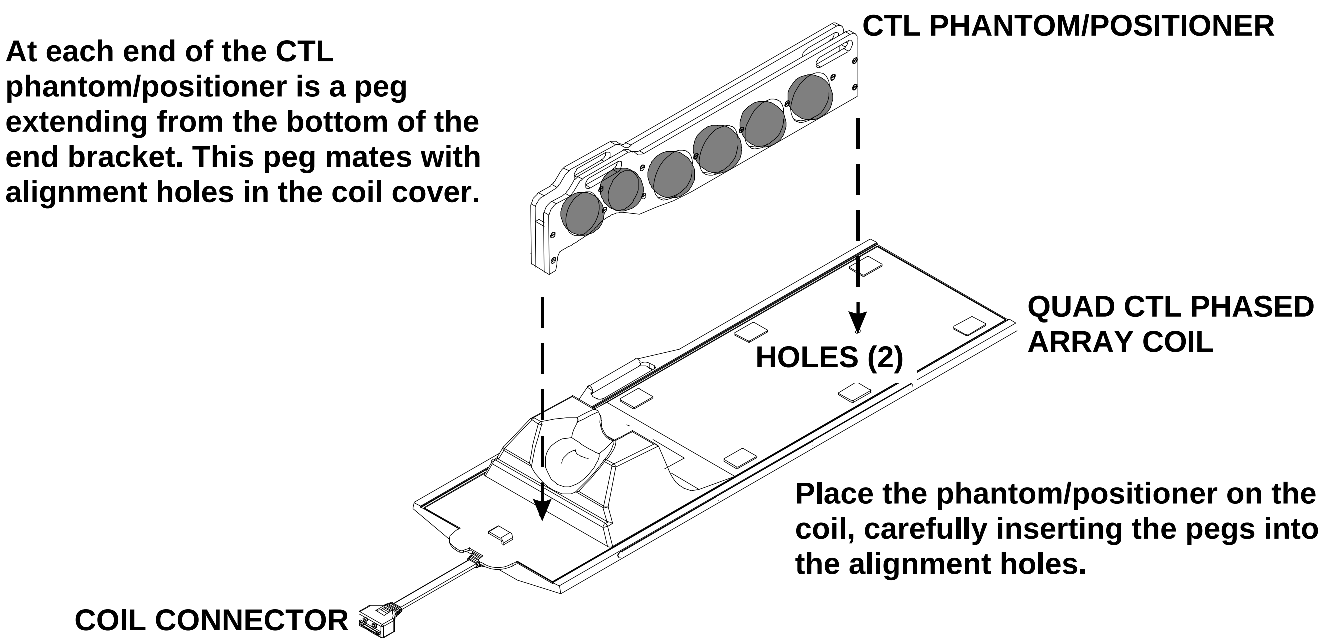
2. Remove Quad Head Coil (if present) from cradle.

3. Place Quad CTL Phased Array Coil to be tested on table.

4. Connect Coil connector to the mating connector in Carriage Assembly.

5. Place the phantom/positioner on the Coil. Ensure the two pegs on the bottom of the positioner mate with the holes on the top cover of the Coil. See Illustration 3-2.

3-2 Quadrature CTL Phased Array Coil Imaging Performance Verification (cont'd)

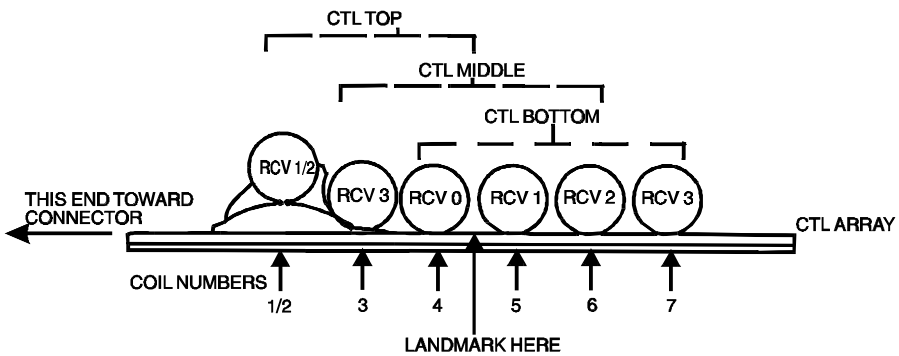


QUAD CTL PHANTOM/POSITIONER SETUP
ILLUSTRATION 3-2

3-2 Quadrature CTL Phased Array Coil Imaging Performance Verification (cont'd)

6. Position the Quad CTL Phased Array Coil and Positioner with phantoms in the center of the cradle. Landmark for center of Coil and advance to isocenter using the [**ADVANCE TO SCAN**] button. See Illustration 3-3.

Note: Testing the CTL Coil requires two scans. Landmark in the center of the coil-See Illustration 3-3. Scan the CTL TOP (S180), this tests the top half of the array. Scan the CTL BOT (I120), this test the bottom half of the array. The phantom spheres are on 120 mm centers.



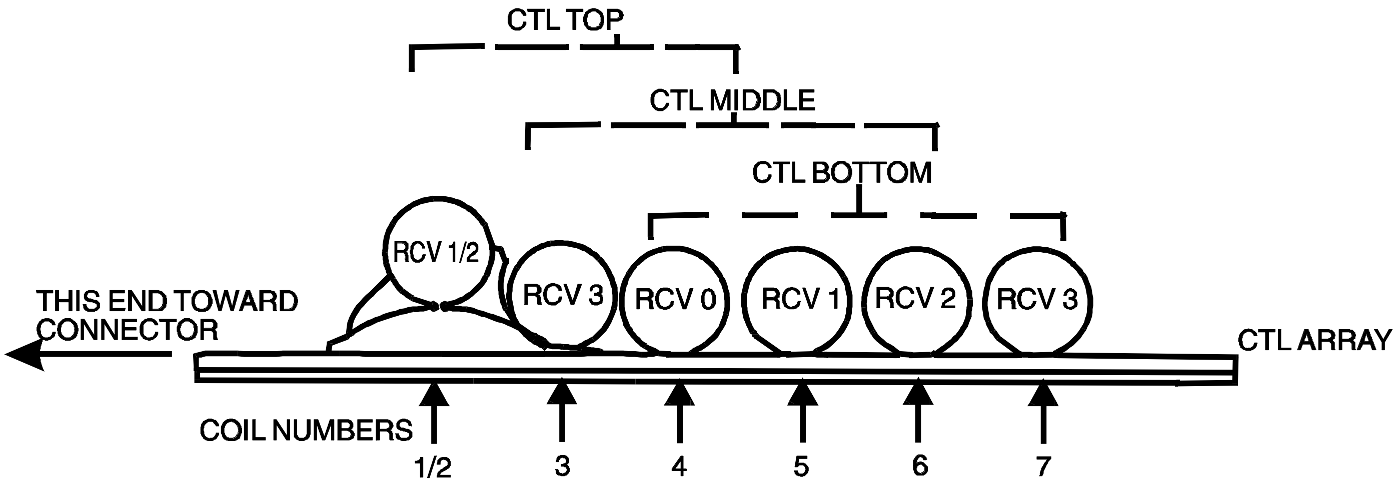
Note: The Quad CTL array contains seven coils; four are used at one time. Two regions are designated. Landmarks are marked on the side of the array. The CTLTOP array uses coils 1 through 4 - Black Marking. CTLBOT uses coils 4 through 7 - Black Marking. If CTL Coil is landmarked as above, Scan Location for CTLTOP is S180 and CTLBOT is I120.

PHANTOM/POSITIONER LANDMARK S
ILLUSTRATION 3-3

3-2 Quadrature CTL Phased Array Coil Imaging Performance Verification (cont'd)

Phased Array Coil Receiver Paths

The Phased Array Coils have multiple individual antenna loop elements inside the assembly; the TPS chassis has multiple receivers (RCV0-RCV3). See Illustration 3-4.



QUAD CTL PHASED ARRAY COIL TO TPS RECEIVER CORRESPONDENCE
ILLUSTRATION 3-4

3-2 Quadrature CTL Phased Array Coil Imaging Performance Verification (cont'd)

7. Setup Scan Prescription as shown in Table 3-2.

TABLE 3-2
Quadrature CTL Phased Array Coil SNR - Scan Protocol

Id:	**geservice**	Rep Time (TR):	**[300 msec]**
Name:	**quad ctl snr**	Auto CF:	**[Peak]**
Patient Weight:	**300**	Field of View:	**[48 cm]**
Patient Entry:	**[Head First]**	Scan Thickness:	**[3 mm]**
Patient Position:	**[Supine]**	Interscan Spacing:	**[Other] 0**
Axial/Sag. Landmark:	**[Sternal Notch]**	Start Loc (L/R): **0** End Loc (L/R):	**0**
Coil Type:	**[Other Coils] [CTLTOP]**	No. of Scan Location:	**1**
Scan Plane:	**[Sagittal]**	FOV Center (P/A):	**0** (S/I) **S180**
Image Mode:	**[2D]**	Acq. Matrix (freq.):	**[256]**
Pulse Sequence:	**[Spin Echo]**	Acq. Matrix (phase)	**[256]**
Imaging Options:	**[None]**	Frequency Direction:	**[S/I]**
or enter PSD Filename		Imaging Time:	**[1 NEX 1:23]**
Number of Echoes:	**[1]**	Contrast:	**[No]**
Echo Time (TE):	**[30 msec]**	Table Delta:	**0 mm**

3-2 Quadrature CTL Phased Array Coil Imaging Performance Verification (cont'd)

8. Select **[Auto Prescan]** to properly calibrate the RF power level for the 90 degree and 180 degree pulses.

9. Select **[Scan]**. Observe the resulting image of the spheres. Ensure that there are no artifacts of any sort in the sphere image. Record the Exam number and Series number for SNR Calculations.

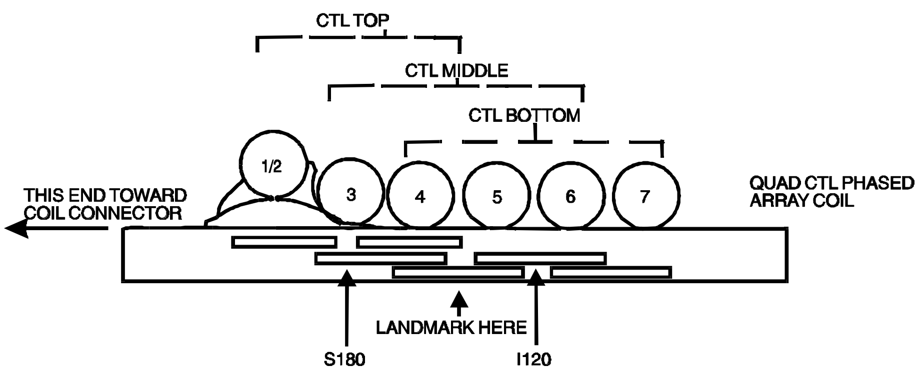
10. Select **[Scan]** again. This second image of the sphere will be used for determination of CTLTOP mode SNR.

11. Select **[Cancel]**. and **[New Series].**

3-2 Quadrature CTL Phased Array Coil Imaging Performance Verification (cont'd)

12. Ensure the landmark is still on center of coil. See Illustration 3-5.

Note: Testing the CTL Coil requires two scans. Landmark in the center of the coil-See Illustration 3-5. Scan the CTL TOP (S180), this tests the top half of the array. Scan the CTL BOT (I120), this test the bottom half of the array. The phantom spheres are on 120 mm centers.



Note: The Quad CTL array contains seven coils; four are used at one time. Two regions are designated. Landmarks are marked on the side of the array. The CTLTOP array uses coils 1 through 4 - Black Marking. CTLBOT uses coils 4 through 7 - Black Marking. If CTL Coil is landmarked as above, Scan Location for CTLTOP is S180 and CTLBOT is I120.

PHANTOM/POSITIONER LANDMARK SETUP CTLBOT
ILLUSTRATION 3-5

3-2 Quadrature CTL Phased Array Coil Imaging Performance Verification (cont'd)

13. Select [**Patient Position**], [**Other Coils**], [**CTLBOT**].

14. Select [**Scanning Range**], FOV center in S/I direction as **I120**, and [**Accept**]. Advance to **SCAN OPERATIONS**.

15. Select [**Auto Prescan**] to calibrate the RF power level for the 90 and 180 degree pulses.

16. Select [**Scan**]. Observe the resulting image of the spheres. Ensure that there are no artifacts of any sort in the sphere images. Record the Exam number and Series number for SNR calculations.

17. Select [**Scan**]. This second image of the spheres will be used for determination of CTLBOT mode SNR.

18. Select [**Cancel**]. Refer to Section 3-3 for SNR image analysis.

3-3 SNR Image Analysis

Description

The SNR Tool retrieves two operator selected images. Signal value is computed as the mean pixel value in a ROI covering 80% of the image. The image is analyzed to determine the center for positioning the ROI. A difference image is created by subtracting the second image from the first and to calculate noise from the subtracted image. Signal value, noise value, and SNR are reported.

3-3-1 SNR Image Analysis for 5.X Systems

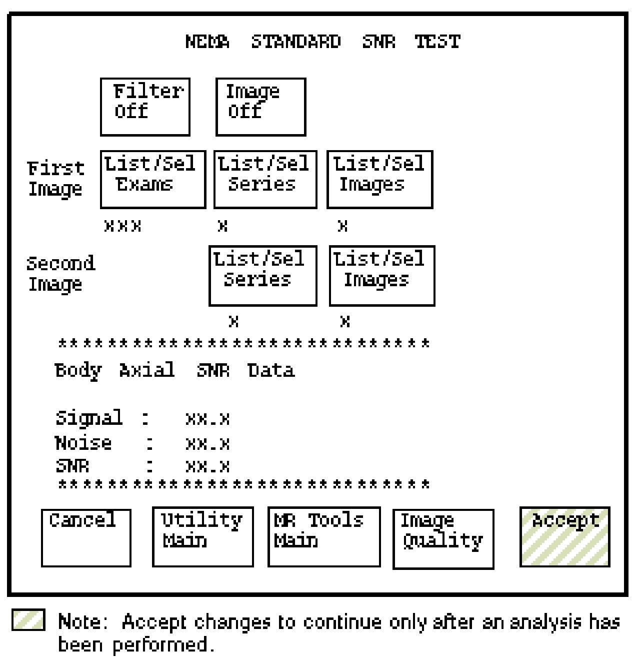
Procedure

1. Touch **[UTILITIES]**, **[MR Tools]**, **[Image Quality]**, then **[SNR Test]**. The SNR Test screen is displayed; see Illustration 3-6.

2. Enter image exam, series, and image numbers. If exam, series, or image numbers are not known, select **[List Exams]**, **[List Series]**, or **[List Images]** to display list to choose from.

Note

Image number selection must be back lit (highlighted) to be able to enter information. Use Switch key on keyboard to transfer control from left to right side of Touch Screen.



SNR TEST SCREEN
ILLUSTRATION 3-6

3-3-1 SNR Image Analysis for 5.X Systems (cont'd)

3. If high pass filtering is desired to be performed on data, touch **[Filter Off]** which will highlight and change to **[Filter On]**.

4. If the difference image annotated with data is to be created, touch **[Image Off]** which will highlight and change to **[Image On]**.

5. Touch **[Accept]** to begin analysis. The final values are displayed on the touch screen, see Illustration 3-6.

6. Touch **[Continue]** then select the next exam and repeat the above analysis for each image pair.

7. Record the date and value calculated in the appropriate column under "SNR Data QA Check" of the Data Table as is instructed.

3-3-2 SNR Image Analysis for 8.X Systems

Description

The SNR Tool retrieves two operator selected images. Signal value is computed as the mean pixel value in a ROI covering 80% of the image. The image is analyzed to determine the center for positioning the ROI. A difference image is created by subtracting the second image from the first and to calculate noise from the subtracted image. Signal value, noise value, and SNR are reported.

Procedure

1. Select **[Service Desktop]**, **[Calibration/Checks]**, then **[Image Quality]**.

<<< NEMA Image Quality Analysis >>>

1. Signal To Noise Check
2. Slice Offset Checks
3. Slice Thickness/Resolution Check
4. T2 Uniformity Check
5. Full Field Distortion Check
6. Exit NEMA test

Select Test: ....**1 [ENTER]**

****
* Select First Image *
****

========================
Image Selection Menu
========================

Current Selection:

Exam_No = 50210, Series_No = 1, Image_No = 1

A. Select Exam
B. Select Series
C. Select Images
D. List/Select Exam
E. List/Select Series
F. List/Select Image
X. Execute the selected test

YOUR CHOICE :..**A [ENTER]** *(Enter appropriate selection, A-F to choose image.)*

Exam No. ?..**50201 [ENTER]** *(Enter*

(Iterate image selection process until Current Selection *is correct.)*

```
=========================
  Image Selection Menu
=========================
```

Current Selection:

Exam_No = 50201, Series_No = 1, Image_No = 1

A. Select Exam
B. Select Series
C. Select Images
D. List/Select Exam
E. List/Select Series
F. List/Select Image
X. Execute the selected test
YOUR CHOICE :
Invalid Entry (Invalid Entry, *always occurs here.*)


```
=========================
  Image Selection Menu
=========================
```

Current Selection:

Exam_No = 50201, Series_No = 2, Image_No = 1

A. Select Exam
B. Select Series
C. Select Images
D. List/Select Exam
E. List/Select Series
F. List/Select Image
X. Execute the selected test
YOUR CHOICE :...**X [ENTER]** (*Perform this step*

```
*************************
*  Select Second Image   *
*************************
```

```
==========================
  Image Selection Menu
==========================
```

Current Selection:

Exam_No = 50210, Series_No = 1, Image_No = 1

A. Select Exam
B. Select Series
C. Select Images
D. List/Select Exam
E. List/Select Series
F. List/Select Image
X. Execute the selected test

YOUR CHOICE :..**A [ENTER]** *(Enter approp selection, A-F to choos image.)*

Exam No. ?..**50201 [ENTER]** *(Enter appropriate numb*

8.x Procedure (cont'd)

(Iterate image selection process until Current Selection *is correct.)*

```
==========================
  Image Selection Menu
==========================
```

Current Selection:

Exam_No = 50201, Series_No = 2, Image_No = 2

A. Select Exam
B. Select Series
C. Select Images
D. List/Select Exam
E. List/Select Series
F. List/Select Image
X. Execute the selected test

YOUR CHOICE :..**X [ENTER]** *(Perform this when* Current Selection *is*

Click left button to go to PREVIOUS ILLUSTRATION, TABLE, or SCREEN

```
**********************************
*       SNR results           *
*                          *
* Signal = 801.040222  Noise = 10.062437  snr = 79.606979 *
**********************************
```

<<< NEMA Image Quality Analysis >>>

1. Signal To Noise Check
2. Slice Offset Checks
3. Slice Thickness/Resolution Check
4. T2 Uniformity Check
5. Full Field Distortion Check
6. Exit NEMA test

Select Test: ....**6 [ENTER]**

2. Record the date and value calculated in the appropriate column under "SNR Data QA Check" of the Data Table (following Section 5, RENEWAL PARTS) as is instructed.

3-4 Checking the Protection PIN Diodes and Output Cable with DMM

CAUTION

As coil tuning and decoupling tuning may be affected by PIN diode replacement, field replacement of the PIN diodes will not be performed.

1. Select the DIODE TEST function on the DMM.

2. Connect the NEGATIVE lead to the the connector pin on row A. Connect the POSITIVE lead to the pin on row B. Refer to Table 3-3 and Illustration 3-7. A reading of 0.500 volts to 0.900 volts should be observed.

3. If a reading below 0.400 volts is observed in either direction, either the output cable is shorted or a bad PIN diode.

4. If a reading above 0.900 is noted, the output cable or the PIN diode is open.

3-4 Checking the Protection PIN Diodes and Output Cable with DMM (cont'd)

5. Connect the NEGATIVE lead to the connector pin on row B. Connect the POSITIVE lead to the connector pin on row A. Refer to Table 3-3.

6. A reading of INFINITY should be observed. If a finite reading is observed in both directions, the output cable or the PIN diodes are shorted or leaky.

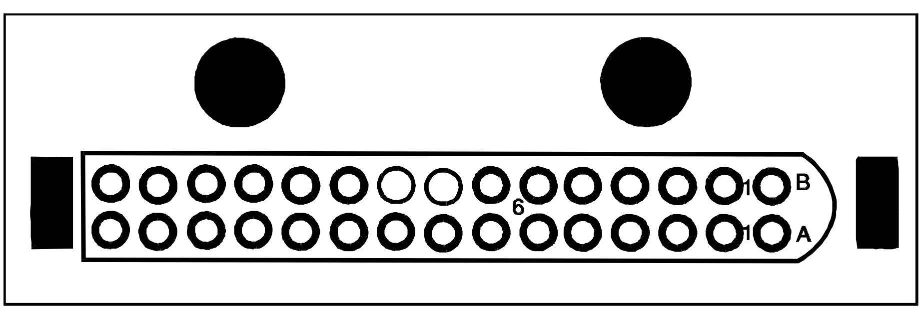
7. To test the cable, perform the above tests while flexing the cable, especially near the 30 pin connector and strain relief exiting the coil.

3-4 Checking the Protection PIN Diodes and Output Cable with DMM (cont'd)

TABLE 3-3

Diode Test Connections

COIL NUMBER	POSITIVE LEAD CONNECTION STEP 2	POSITIVE LEAD CONNECTION STEP 6	NEGATIVE LEAD CONNECTION STEP 2	NEGATIVE LEAD CONNECTION STEP 6
1	3A	4A	4B	3B
2	5A	4A	4B	5B
3	7A	8A	8B	7B
4	9A	8A	8B	9B
5	11A	12A	12B	11B
6	13A	12A	12B	13B
7	15A	14A	14B	15B



PIN LOCATIONS FOR PHASED ARRAY CONNECTOR

ILLUSTRATION 3-7

3-5 Checking the Decoupling Diodes with a DMM

CAUTION

As coil tuning and decoupling tuning may be affected by PIN diode replacement, field replacement of the PIN diodes will not be performed.

1. Remove the bottom cover of the coil. See Section 4-1.

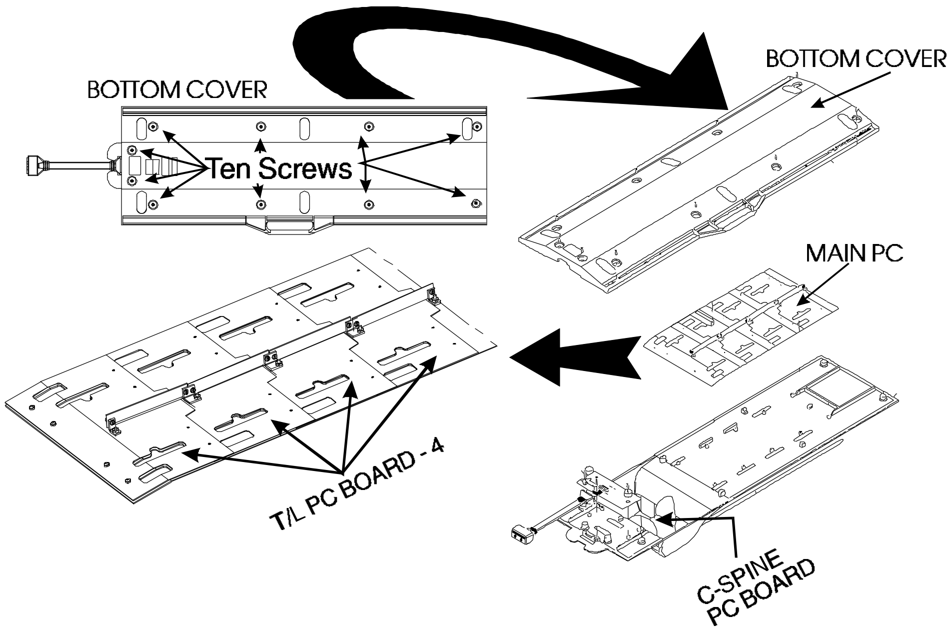
2. Locate the 22 sets of decoupling diodes; four on each of the four T/L PC boards, Four on the inferior C-Spine, and one set in each loop of the superior C-Spine PC board. See Illustration 3-8.

3. Measure across each pair of diodes in both directions with a DMM set to the diode test function. The reading should be 0.400 volts to 0.900 volts in each direction. A lower reading indicates a shorted or leaky diode. A higher reading indicates an open diode.

4. If any of the diode pairs have failed, the coil should be replaced.

3-5 Checking the Decoupling Diodes with a DMM (cont'd)

5. Attach the Bottom Cover. See Section 4-1. Perform the performance tests in Section 3-2 - Quad CTL Phased Array Coil SNR Verification.



PROTECTION DIODES LOCATIONS
ILLUSTRATION 3-8

4-1 Disassembly / Assembly of 1.0T Quad CTL Phased Array Coil

Note:

Each antenna loop element is tuned at the factory to provide the proper input impedance. None of these boards are field replaceable. These circuits must remain together as a matched set.

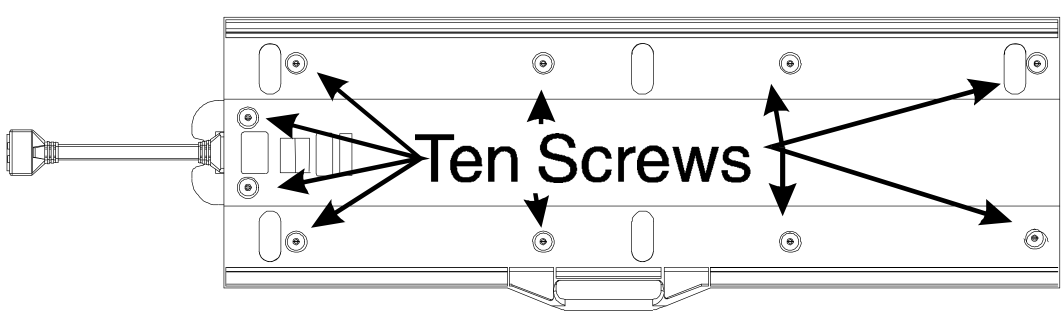
1. Remove the ten screws holding the top and bottom housing together. See Illustration 4-1.

2. Separate the two housings from each other.

3. To reassemble, put the two housings together and **lightly** tighten the ten screws.

Note:
The screws are very susceptible to breakage from overtightening.

4-1 Disassembly / Assembly of 1.0T Quad CTL Phased Array Coil (cont'd)



DISASSEMBLY/ASSEMBLY OF QUAD CTL PHASED ARRAY COIL
ILLUSTRATION 4-1

4-2 Replacing the External Cable.

1. Disassemble the top and bottom housings. Refer to Section 4-1.

2. Remove the eight SMA connectors on the External Cable from the mating connectors on the Phase Compensation Board. Remove the two screws and the Strain Relief holding the External Cable in place. Remove the External Cable.

3. Install the new External Cable in place, ensuring that the Strain Relief is placed in the same direction as the original. See Illustration 4-2.

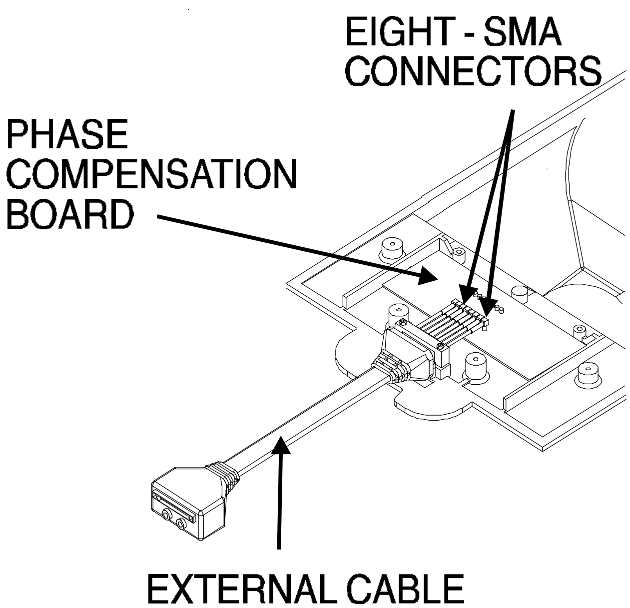
4. Install the Strain Relief and screws in place.

5. Reconnect the eight SMA connectors to the Phase Adjustment Board in the proper order.

6. Assemble the top and bottom housings together. Refer to Section 4-1.

7. Repeat the Performance Tests in Section 2-4.

4-2 Replacing the External Cable. (cont'd)



REPLACEMENT OF EXTERNAL CABLE
ILLUSTRATION 4-2

4-3 Replacing the Protection (PIN) Diodes

CAUTION

As coil tuning and decoupling tuning may be affected by PIN diode replacement, field replacement of the PIN diode will not be performed.

1. PIN Diodes are not field replaceable. Return the coil for replacement.

4-4 Cleaning the Coil

Caution

Avoid damaging sensitive electronic parts. Do not spray or pour dishwashing solution directly onto the Quadrature CTL Phased Array Coil, or external cable. Do not use alcohol to clean the Quadrature CTL Phased Array Coil. Never submerge the Quadrature CTL Phased Array Coil in any liquid.

Clean the Quadrature CTL Phased Array Coil and external cable with a mild dishwashing liquid and water solution. Wet a soft cloth with the solution and proceed to clean.

Before returning the coil for servicing, thoroughly clean the coil, cable, and connectors with a 10% bleach in water solution.

Section 5 - Renewal Parts

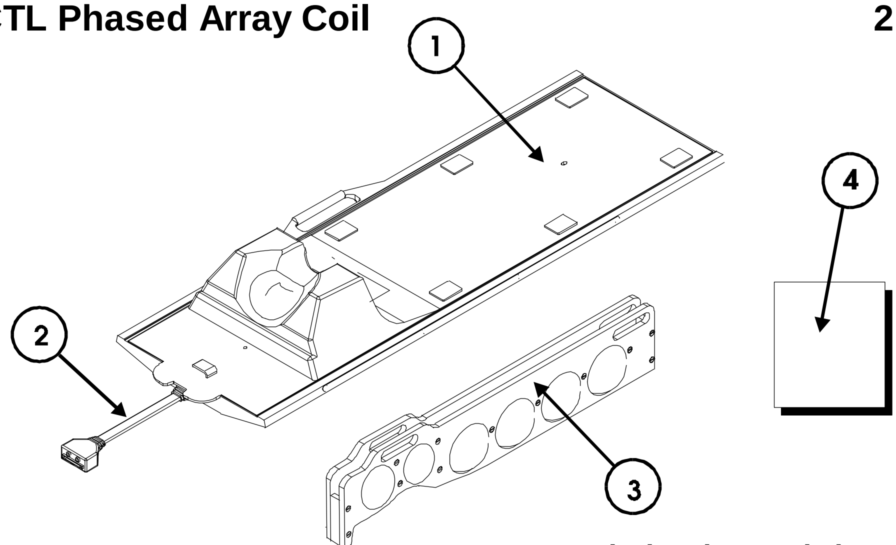
WARNING

BIO HAZARD! EQUIPMENT BEING RETURNED FROM USE IN A CLINICAL SETTING MUST BE CLEAN AND FREE OF BLOOD AND OTHER INFECTIOUS SUBSTANCES. THE DEPARTMENT OF TRANSPORTATION (DOT) HAS RULED THAT ITEMS THAT WERE SATURATED AND/OR DRIPPING WITH HUMAN BLOOD THAT ARE NOW CAKED WITH DRIED BLOOD; OR WHICH WERE USED OR INTENDED FOR USE IN PATIENT CARE ARE REGULATED MEDICAL WASTE FOR TRANSPORTATION PURPOSES AND MUST BE TRANSPORTED AS A HAZARDOUS MATERIAL. UNDER NO CIRCUMSTANCES SHOULD A PART OR EQUIPMENT WITH VISIBLE BODY FLUIDS BE TAKEN OR SHIPPED FROM A CLINIC OR SITE (FOR EXAMPLE, SURFACE COILS).

1. Employees shall follow proper decontamination procedures for clean up of bloodborne pathogens. Refer to Section 4-4, Cleaning The Coil. It is the responsibility of the GEMS employee to insure the part/equipment has been properly decontaminated prior to shipment.

Section 5 - Renewal Parts - cont'd

1.0T Quadrature CTL Phased Array Coil 2151177



Item	Part Number	FRU	Name	Qty.	Description (Remarks)
1	2151178-4	1	Quad CTL	1	1.0T Quad CTL Phased Array Coil
2	2100937-15	1	Cable	1	External Cable
3	2100937-16	1	Holder	1	Quad CTL Phantom Holder
4	46-328040G6	1	Preamp	2	1.0T Multicoil Preamp Asm.

Data Table

Use the space provided below to record the calculated signal to noise ratio (SNR) data obtained from Section 3, Functional Checks. After recording the SNR data obtained during the initial coil installation, record subsequent SNR data in column 2 - 4 below and the date they were obtained in column number 1 as a periodic QA check. If the ratio of any of the coils found is not greater than 85%, then there is a problem in the coil or the MR system.

Original SNR data obtained at initial coil installation:

Body Coil SNR Value: ________________

Quadrature CTL Phased Array Coil SNR Value: CTLTOP ________
CTLBOT ________

Date: ______________

Data Table (continued)

SNR Data QA (Quality Assurance) Check:

1	2	3	4	
Date:	**CTLTOP**	**CTLBOT**	**Are new Values Divided by Original Values > 85%?**	
________	________	________	________	Y/N
________	________	________	________	Y/N
________	________	________	________	Y/N
________	________	________	________	Y/N
________	________	________	________	Y/N
________	________	________	________	Y/N
________	________	________	________	Y/N
________	________	________	________	Y/N
________	________	________	________	Y/N
________	________	________	________	Y/N
________	________	________	________	Y/N

Make additional copies of this document as is needed.

Defective Coil Return Form

Note

To allow for proper assessment of defective returned coils, this form must be completely filled out and accompany all returned coils. Include films or prints of any image quality related complaints with a description of scan protocol used.

Date____
Site Name____
Site Address____
Service Engineer____
Coil Serial Number____
Date Coil Installed____
Description of Coil Problem____________________________________
____
____
____
____
____

Defective Coil Return Form (continued)

Note

An alternate proprietary procedure is available for GE use and to customers with a valid Advanced Service Package Limited License. Refer to "TLT PROCEDURE" located on *CD-ROM Direction 15520, navigate to System: Troubleshooting*. Use Table DR-1 and DR-2 to record the TLT results of the defective coil.

TABLE DR-1
TLT DATA FOR DEFECTIVE SURFACE COIL

SITE: ______________________________**NAME:** _____________________**DATE:** ___________
TLT FILE NUMBER: _____________________________
SNR: NOISE ______________________________
SNR MEAN ______________________________
SNR SIGNAL ______________________________
SNR AREA ______________________________
TR MAP: 89-91 ____________________________%
85-95 ____________________________%
65-115 ____________________________%
FLIP MEAN ______________________________

Defective Coil Return Form (continued)

TABLE DR-2
TLT DATA FOR DEFECTIVE PHASED ARRAY COIL

SITE: ______________________________ **NAME:** ______________________ **DATE:** ___________
TLT FILE NUMBER: ______________________________
SNR: NOISE (0) ____________________________
NOISE (1) ____________________________
NOISE (2) ____________________________
NOISE (3) ____________________________
NOISE (C) ____________________________
SNR MEAN (0) ____________________________
SNR MEAN (1) ____________________________
SNR MEAN (2) ____________________________
SNR MEAN (3) ____________________________
SNR MEAN (C) ____________________________
TR MAP: 89-91 (C) _________________________%
(Combined 85-95 (C) _________________________%
Results) 65-115 (C) ___________________________%
FLIP MEAN (C) ____________________________
FLIP SDV (C) ____________________________
RCV MEAN (C) ____________________________
RCV SDV (C) ____________________________

Damage in Transportation Statement

<u>All packages should be closely examined at time of delivery.</u> If damage is apparent, have notation **"damage in shipment"** written on **all** copies of the freight or express bill **<u>before</u>** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

Call Traffic and Transportation, Milwaukee, WI (414) 785-5052/8*323-5052 **immediately** after damage is found. At this time be ready to supply name of carrier, delivery date, consignee name, freight or express bill number, item damaged and extent of damage.

Complete instructions regarding claim procedure are found in Section "S" of the Policy & Procedure Bulletins.

3/12/92

GE Medical Systems

GE Medical Systems: Telex 3797371
P.O. Box 414, Milwaukee, Wisconsin 53201 U.S.A.
(Asia, Pacific, Latin America, North America)

GE Medical Systems — Europe: Telex 261794
Shortlands, Hammersmith, London W6 8BX U.K.

Note Page (optional)

NOTES -- This page is used to keep all of the notes for this procedure.
(Optional)

Release Notes (required)

RELEASE NOTES -- This page is used to hold all release notes for this procedure (required)

This is the first release of this procedure in Toolbook. There are no notes.

Initial Comments (required)

INITIAL COMMENTS -- This page is used to hold all initial comments for a procedure. (Required)

Revision 0 ..... January, 1996 ..... Initial Release.
Revision 1 ..... October, 1997 ..... Add Multilingual, add Bloodborne pathogen warnings, add 8.x SNR verification test, misc control number changes.

Revision Information (required)

REVISION INFORMATION - this is for revision information for a procedure. (Required)

Revision#: 0
Date: 01/04/96
Class: A
87305-T-286
Author: George E. Lang
Title: Signa® 1.0T Quadrature CTL Phased Array Coil
Revision Description:

Revision#: 1
Date: 10/22/97
Class: A
87305-T-286 Rev. A
Author: George E. Lang
Title: Signa® 1.0T Quadrature CTL Phased Array Coil
Revision Description:Add Multilingual, add Bloodborne pathogen warnings, add 8.x SNR verification test, misc control number changes.

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